

TRAUMATIC CARDIAC ARREST

DO NOT START TREATMENT IF THERE ARE SIGNS OF CEREBRAL DESTRUCTION, DECAPITATION OR HEMICORPECTOMY. DO NOT PRIORITISE CHEST COMPRESSIONS IN THE TRAUMATIC ARREST.

PATIENT IN CARDIAC ARREST or AGONAL STATE

PENETRATING TRAUMA TO THORAX,
OBVIOUS SIGNIFICANT HAEMORRHAGE or
BLUNT TRAUMATIC INJURIES

LOW ENERGY MECHANISM or NO OBVIOUS
TRAUMATIC INJURIES/AETIOLOGY

COMMENCE HOTT APPROACH

COMMENCE MEDICAL CARDIAC ARREST
MANAGEMENT

H

HYPOVOLAEMIA

MASSIVE HAEMORRHAGE CONTROL

APPLY TOURNIQUETS TO OBVIOUS OPEN LIMB INJURIES
APPLY PELVIC BINDER IN BLUNT POLYTRAUMATIC INJURIES

VOLUME RESUSCITATION

GAIN IV/IO ACCESS (TWO ACCESS POINTS IF POSSIBLE)
RAPIDLY INFUSE WARMED BLOOD or 500ml FLUID BOLUSES & TXA 1g

O

OXYGENATE

AIRWAY MANAGEMENT

IMMEDIATE ADVANCED AIRWAY PLACEMENT WITH SGA
CONSIDER SURGICAL AIRWAY IN SEVERE FACIAL BURNS

MAXIMISE OXYGENATION

HIGH FLOW OXYGEN THROUGH AIRWAY CIRCUIT WITH ETCO₂
CONNECTED (<1.0kPa IS POOR INDICATOR)
DIFFICULTY TO VENTILATE = ↑ INTRATHORACIC PRESSURE
PPV CAN CAUSE TENSION PATHOLOGY

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TENSION

RESPIRATORY

PERFORM FINGER/OPEN THORACOSTOMIES IN PATIENTS WITH CHEST
INJURIES AND NOTE DIFFICULTY IN VENTILATION
NEEDLE DECOMPRESSION CAN BE DONE IF NOT TRAINED IN ABOVE
CHECK FOR HOLES IN THORAX IF POSSIBLE

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TAMPONADE

CIRCULATORY COLLAPSE (<15 MINUTES)

RESUSCITATIVE THORACOTOMY INDICATIONS (Page 25):
IF NO RESPONSE TO ABOVE MANAGEMENT OR PENETRATING TRAUMA TO
THORACIC/UPPER ABDOMINAL REGION
USED TO RELIEVE CARDIAC TAMPONADE OR APPLY AORTIC PRESSURE

CONSIDER CHEST COMPRESSIONS BUT ONLY IF HYPOVOLAEMIA IS CORRECTED AND THERE ARE NO SIGNIFICANT CHEST INJURIES. DEFIBRILLATION IS RARELY NEEDED.

ADRENALINE (1mg 1:10,000) CAN BE ADMINISTERED EVERY 3-5 MINUTES IF THERE IS NO RESPONSE TO VOLUME RESUS.

NEUROGENIC SHOCK (FILL AND SQUEEZE APPROACH)

TCA IS HIGHLY LIKELY TO BE FROM HYPOVOLAEMIA, HOWEVER, NEUROGENIC SHOCK CAN BE A CONCOMITANT FACTOR THAT NEEDS TO BE ADDRESSED AND MIGHT NOT RESPOND TO VOLUME. IF THERE WAS EVIDENCE OF SPINAL CORD INJURY PERI-ARREST (BRADYCARDIA, HYPOTENSION, WARM PERIPHERIES, DISTENDED PERIPHERAL VEINS) THEN ADRENALINE WOULD BE BENEFICIAL, HOWEVER, HYPOVOLAEMIA SHOULD BE ADDRESSED FIRST WITH VOLUME RESUS. CONSIDER ADRENALINE INFUSION OF 2 - 10mcg/min IF ROSC IS ACHIEVED FOR THESE PATIENTS.

CONSIDER POINT OF CARE ULTRASOUND (POCUS)

POCUS CAN BE USEFUL TO IDENTIFY POSSIBLE INTRATHORACIC BLEED (PRESENCE OF FREE FLUID), CARDIAC TAMPONADE (SWINGING HEART, RV COMPRESSION) AND TO IDENTIFY ANY CARDIAC MOVEMENT.

	HEART	IVC	LUNG	ABDOMEN
HYPOVOLAEMIA	SMALL & HYPER-KINETIC CHAMBERS	SLIGHT	NO B LINES FREE FLUID	FREE FLUID AAA
CARDIAC TAMPONADE	PERICARDIAL EFFUSION RV COMPRESSION	DILATED	NO B LINES	N/A
TENSION PNEUMOTHORAX	SMALL & HYPER-KINETIC CHAMBERS	DILATED	ABSENCE OF LUNG SLIDING	N/A

RETURN OF SPONTANEOUS CIRCULATION (ROSC)

AIRWAY	REASSESS AIRWAY AND CONSIDER POST-ROSC SEDATION AND UPGRADING TO A DEFINITIVE AIRWAY IN PROLONGED CARE. CONTINUE VOLUME RESUSCITATION.
RESPIRATORY	CONTINUE VENTILATION AND OXYGENATION WITH ET _{CO} ₂ CONNECTED. AVOID PPV INDUCED RAISED INTRATHORACIC PRESSURE = CIRCULATORY COLLAPSE
CIRCULATORY	CONTINUE RESUSCITATION WITH BLOOD PRODUCTS IF POSSIBLE TO RADIAL PULSES OR MAP >65mmHg. FLUIDS CAN BE ADMINISTERED IN 250ml BOLUSES THEN REASSESS. ADDRESS TRAUMA INDUCED COAGULOPATHY CONCERNS IF POSSIBLE. CONSIDER INOTROPES/VASOPRESSORS: METARAMINOL 1ml BOLUS THEN 6-10ml/hr INFUSION ADRENALINE 2-10mcg/min (1:100,000 0.2-1ml/min TITRATING TO MAP >70mmHg)
HEAD (DISABILITY)	CONSIDER SEDATION FOR NEUROPROTECTION BUT ENSURE THAT PATIENT HAS BEEN OPTIMISED WITH VOLUME RESUSCITATION PRIOR.
EXTENSIVE INJURIES	IT IS REASONABLE TO PERFORM SEDATION IN PATIENTS WITH EXTENSIVE INJURIES TO AVOID UNNECESSARY SUFFERING, ESPECIALLY IF SURGERY IS REQUIRED AND LIKELY TO BE PERFORMED IN A REASONABLE TIME-FRAME. VOLUME RESUSCITATION MUST STILL TAKE PLACE IN THIS INSTANCE TO AVOID DRUG INDUCED ADVERSED EVENTS.

PATIENT REQUIRES EMERGENCY TRANSFER TO DAMAGE CONTROL SURGERY WHILST CONTINUING DAMAGE CONTROL RESUSCITATION.

WHEN TO TERMINATE RESUSCITATION

ALL EFFORTS HAVE BEEN EXHAUSTED WITH NO RESPONSE TO TREATMENTS
 PROLONGED ASYSTOLE, PERSISTENTLY LOW ET_{CO}₂ AND CARDIAC STANDSTILL SEEN ON POCUS
 LIFE-SAVING RESOURCES ARE NEEDED ELSEWHERE IN MULTIPLE CASUALTIES
 DO NOT START TREATMENT IF THERE ARE SIGNS OF CEREBRAL DESTRUCTION, DECAPITATION, HEMICORPECTOMY OR SIGNIFICANT NON-SURVIVABLE INJURIES.

Guideline written in accordance with SFM Clinical Standard Operating Guideline 009 'Traumatic Cardiac Arrest' (2022) and MERT Standard Operating Procedure 'Traumatic Cardiac Arrest V3.1' (2022).

References

Barnard, E., Hunt, P., Lewis, P., Smith, J.E. 2018. The Outcome of Patients in Traumatic Cardiac Arrest Presenting to Deployed Military Medical Treatment Facilities: Data from the UK Joint Theatre Trauma Registry. *J R Army Med Corps*, 164(3), pp.150-154.

Faculty of Pre-hospital Care. 2018. Consensus Statement: Management of Traumatic Cardiac Arrest. Available at: [Microsoft Word - TCA submission Oct 2018.docx \(rcsed.ac.uk\)](#) (Accessed 19 March 2024).